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Ten Practical Rules for Creating Effective Charts, Figures, and Visuals Using SAS®

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ABSTRACT

Effective charts, figures, and visuals are essential for communicating data clearly and accurately, yet their creation is rarely straightforward or automatic. The same dataset can be represented in many ways, such as histograms, scatter plots, bar charts, or pie charts, and even when the same visualization type is used, interpretation can vary widely across audiences. A more inclusive perspective views data visualization as a graphical interface between people and data, emphasizing clarity, accessibility, and intent. This paper builds on the work of Nicolas P. Rougier, Michael Droettboom, and Philip E. Bourne by presenting ten practical rules for improving the design and production of charts, figures, and visuals using SAS®.

INTRODUCTION

The familiar idiom, “A picture is worth a thousand words,” has taken on renewed significance in the digital age, where data is increasingly communicated through visual means. Yet, representing data effectively is not a simple or uniform task. The same dataset can be displayed using a variety of formats, histograms, scatter plots, line charts, bar charts, and pie charts, and even identical visualizations may be interpreted differently depending on the audience. A more precise view of data visualization is as a graphical interface between people and data, where design choices directly influence understanding. This paper presents a set of foundational principles for creating effective charts, figures, and visuals, along with common pitfalls to avoid.

Because patterns, trends, and relationships are not always apparent in tables or reports, visualization techniques offer a powerful advantage for communicating insights. Well-designed visuals can enhance comprehension and support more informed decision-making for data analysts, statisticians, and other professionals. The SAS user community benefits from a robust suite of tools for producing high-quality graphical output to support data exploration, analysis, and statistical modeling. This paper introduces ten practical rules for leveraging SAS ODS Statistical Graphics, its features, capabilities, and syntax, to create clearer, more effective charts, figures, and visuals.

DATASET USED IN EXAMPLES

Each example illustrated in this paper uses a Movies dataset (table) consisting of six columns: title, length, category, year, studio, and rating. Title, category, studio, and rating are defined as character columns with length and year being defined as numeric columns, shown below.

	Title	Length	Category	Year	Studio	Rating
1	Brave Heart	177	Action Adventure	1995	Paramount Pictures	R
2	Casablanca	103	Drama	1942	MGM / UA	PG
3	Christmas Vacation	97	Comedy	1989	Warner Brothers	PG-13
4	Coming to America	116	Comedy	1988	Paramount Pictures	R
5	Dracula	130	Horror	1993	Columbia TriStar	R
6	Dressed to Kill	105	Drama Mysteries	1980	Filmways Pictures	R
7	Forrest Gump	142	Drama	1994	Paramount Pictures	PG-13
8	Ghost	127	Drama Romance	1990	Paramount Pictures	PG-13
9	Jaws	125	Action Adventure	1975	Universal Studios	PG
10	Jurassic Park	127	Action	1993	Universal Pictures	PG-13
11	Lethal Weapon	110	Action Cops & Robber	1987	Warner Brothers	R
12	Michael	106	Drama	1997	Warner Brothers	PG-13
13	National Lampoon's Vacation	98	Comedy	1983	Warner Brothers	PG-13
14	Poltergeist	115	Horror	1982	MGM / UA	PG
15	Rocky	120	Action Adventure	1976	MGM / UA	PG
16	Scarface	170	Action Cops & Robber	1983	Universal Studios	R
17	Silence of the Lambs	118	Drama Suspense	1991	Orion	R
18	Star Wars	124	Action Sci-Fi	1977	Lucas Film Ltd	PG
19	The Hunt for Red October	135	Action Adventure	1989	Paramount Pictures	PG
20	The Terminator	108	Action Sci-Fi	1984	Live Entertainment	R
21	The Wizard of Oz	101	Adventure	1939	MGM / UA	G
22	Titanic	194	Drama Romance	1997	Paramount Pictures	PG-13

TEN SIMPLE RULES FOR BETTER FIGURES

In Rougier, Droettboom and Bourne's published paper, "Ten Simple Rules for Better Figures," a set of rules to improve figure design and to explain common pitfalls associated with the production of better figures are presented, below.

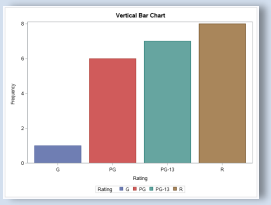
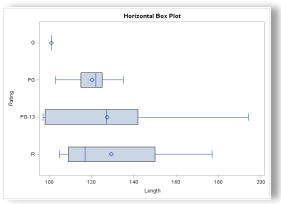
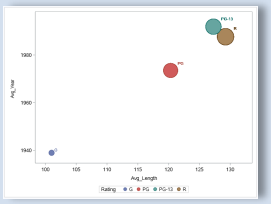
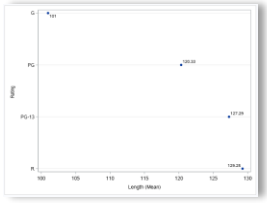
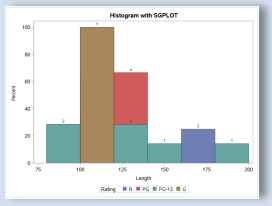
Rule	Description
Rule #1: **Know Your Audience**	When publishing a figure, make sure you explain the concept and verify that your figure is correct and conveys all the relevant information to a broader audience.
Rule 2: **Identify Your Message**	It is important to clearly identify the role of the figure describing the underlying message by determining whether a figure can best express this message.
Rule 3: **Adapt the Figure to the Support Medium**	A figure can be displayed on a variety of media, such as a poster, a computer monitor, a projection screen, or a sheet of paper (as in a printed article). Each of these media represents different physical sizes for the figure, but more importantly, each implies different ways of viewing and interacting with a figure. It is important to keep in mind that during oral presentations, figures will be video-projected and will be seen from a distance, and figure elements must consequently be made thicker (lines) or bigger (points, text), colors should have strong contrast, and vertical text should be avoided, etc.
Rule 4: **Captions Are Not Optional**	A caption explains how to read the figure and provides additional precision for what cannot be graphically represented. This can be thought of as the explanation you would give during an oral presentation, or in front of a poster, but with the difference that you must think in advance about the questions people would ask.
Rule 5: **Do Not Trust the Defaults**	Plotting libraries and/or software comes with a set of default settings. When the end-user does not specify anything, these default settings are used to specify size, font, colors, styles, ticks, markers, etc. Since these settings are to be used for virtually any type of plot, they are not fine-tuned for a specific type of plot. In other words, they are good enough for any plot but they are best for none. All plots require at least some manual tuning of the different settings to better express the message, be it for making a precise plot more salient to a broad audience, or to choose the best color-map for the nature of the data.
Rule 6: **Use Color Effectively**	Color is an important dimension in human vision and is consequently equally important in the design of a figure. However, as explained by Edward Tufte, color can be either your greatest ally or your worst enemy if not used properly. If you decide to use color, you should consider which colors to use and where to use them. For example, you can use color to highlight some element of a figure while keeping other elements gray or black.
Rule 7: **Do Not Mislead the Reader**	A figure is, by definition, tied to the data (be it an experimental setup, a model, or some results) and if you loosen this tie, you may unintentionally project a different message than intended. Avoid making explicit choices that are wrong by design, such as using pie charts or 3-D charts to compare quantities. As a rule of thumb, make sure to always use the simplest type of plots that can convey your message and make sure to use labels, ticks, title, and the full range of values when relevant.
Rule 8: **Avoid "Chartjunk"**	Chartjunk refers to all the unnecessary or confusing visual elements found in a figure that do not improve the message (in the best case) or add confusion (in the worst case). The term was first coined by Edward Tutfe, in which he describes that any decorations that do not tell the viewer something new must be banned.
Rule 9: **Message Trumps Beauty**	If an article displays a figure to introduce results, you might want to try to adapt the figure for your own needs (note to be careful with image copyright). A message and readability of the figure is the most important aspect while beauty is only an option.
Rule 10: **Use the Best Tool**	It is important to recognize that the software and/or library you're using to construct a visual may be different from the software and/or library you're using to conduct your research and/or analyze your data. Whether drawing a graph, designing a schema of your experiment, or plotting data, there are SAS procedures and open-source tools available.

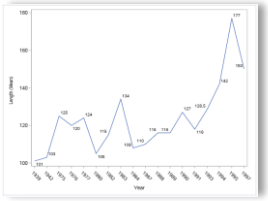
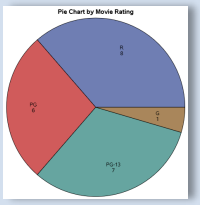
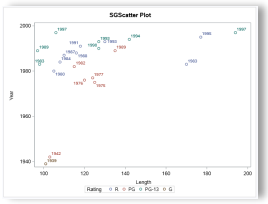
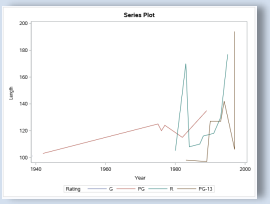
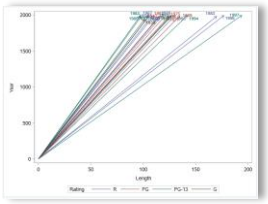
GRAPHICAL DESIGN PRINCIPLES

Good graphical design begins with displaying data clearly and accurately. Data (and information) should be conveyed effectively and without ambiguity. Unnecessary information often distracts from the message; therefore, it should be excluded. In their highly acclaimed book, Statistical Graphics Procedures by Example (2011), Mantange and Heath share this message about graphical output, *"A graph is considered effective if it conveys the intended information in a way that can be understood quickly and without ambiguity by most consumers."*

THE SGPLOT PROCEDURE PLOT TYPE, VISUAL AND DESCRIPTION

Whether the data you work with is large or small, or some size in between, the SGPLOT procedure is a powerful tool for handling many of your graphical needs. You'll be able to let your visuals do the talking by helping your audience see hidden, or hard to see, things in your data, while avoiding the obvious by surprising and engaging your audience. PROC SGPLOT creates single-cell bar charts, box plots, bubble plots, dot plots, histograms, line plots, scatter plots, and an assortment of other plot types quickly and easily. The SGPLOT procedure supports the following plot types and output results.

Plot Type	Visual	Description
Bar		**Bar charts show the relationships among variables and the distribution of a categorical variable as horizontal (HBAR) or vertical (VBAR) output.**
Box		**Box charts are often used during data analysis for the purpose of showing the shape of a distribution, its central value, and its variability.**
Bubble		**Bubble plots produce three-dimensional output to help visualize data and show the relationship between categorized circles (bubbles).**
Dot		**Dot plots provide an effective way to compare frequency counts within categories or groups of values.**
Histogram		**Histograms provide an effective way to view continuous data where the categories represent ranges of data.**

Line		**Line plots provide a quick and easy way to show and organize the frequency of data as horizontal (HLINE) or vertical (VLINE) output.**
Pie		**Pie charts illustrate data proportionality by displaying the quantities of one or more (maximum of six) categories in the form of pie slices.**
Scatter		**Scatter plots are most effectively used to show how much one variable is affected by another.**
Series		**Series plots provide a convenient way to visualize numeric response data over a continuous axis.**
Vector		**Vector charts provide a visual of the x and y values along with the length and direction of data values.**

BEST PRACTICE GRAPHICAL TECHNIQUES

Best practice graphical techniques emphasize essential visual elements in a graph or chart for maximum effectiveness. The following best practice graphical techniques are valuable considerations as you design and develop graphical output and results.

- ✓ Group similar elements together to differentiate them from other elements
- ✓ Use distinct colors to set elements apart
- ✓ Use different shapes to distinguish points on a line graph
- ✓ Varying 2-dimensional elements in a line graph offers an effective way to indicate differences, as well as how much difference exists
- ✓ Length serves as an excellent way to communicate differences between elements
- ✓ Size can show that differences exist but doesn't show the amount of difference between elements
- ✓ Width is good for showing that differences exist but fails to show the actual difference between elements
- ✓ Color saturation is useful for showing distinctions and differences between elements.

Tips for Designing Effective Bar Graphs

- ✓ Avoid misleading scales by starting quantities at zero
- ✓ Verify that the vertical axis is about 25% shorter than the horizontal axis
- ✓ Verify that bars are all the same width
- ✓ Verify that bars are arranged in a logical sequence
- ✓ Use tick marks to display amounts
- ✓ Verify that the space between bars is about half as wide as the bar itself.

Tips for Designing Effective Line Graphs

- ✓ Avoid misleading scales by starting quantities at zero
- ✓ Verify that the vertical axis is about 25% shorter than the horizontal axis
- ✓ Use grid lines to display quantities and amounts.

Tips for Designing Effective Pie Charts

- ✓ Limit the number of pie slices to no more than six
- ✓ Specify the largest pie slice at the top and work clockwise in decreasing order
- ✓ Combine “small” slices into a “Miscellaneous” or “Other” slice
- ✓ Label each pie slice inside the pie slice
- ✓ Emphasize a particular pie slice by color or by extracting it from the pie itself.

CREATING VISUALS WITH STATISTICAL GRAPHIC PROCEDURES

With the availability of ODS Statistical Graphics the SAS user community has a powerful set of tools to produce high-quality graphical output for data exploration, data analysis, and statistical analysis. All that is required to leverage the power available in ODS Statistical Graphics is a Base SAS® license. With that, ODS Statistical Graphics and its many capabilities are available to SAS software users in the form three procedures: SGPLOT, SGSCATTER and SGPANEL.

THE SGPLOT PROCEDURE BY EXAMPLE

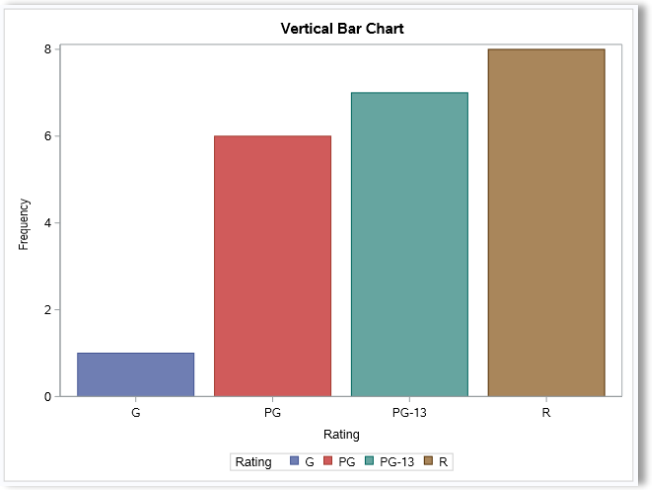
The syntax for the SGPLOT procedure is displayed below.

```
PROC SGPLOT < DATA=data-set-name > < options > ;
  Plot-statement(s) plot-request-parameters < / options > ;
RUN ;
```

Creating a Bar Chart

Bar charts are a commonly used graph type for displaying and comparing the quantities, frequencies or other measurements of discrete categories or groups. The selected variable is displayed along the horizontal (x-axis) and the frequency along the vertical (y-axis). The next SGPLOT displays the distinct values for the RATING variable using a VBAR statement.

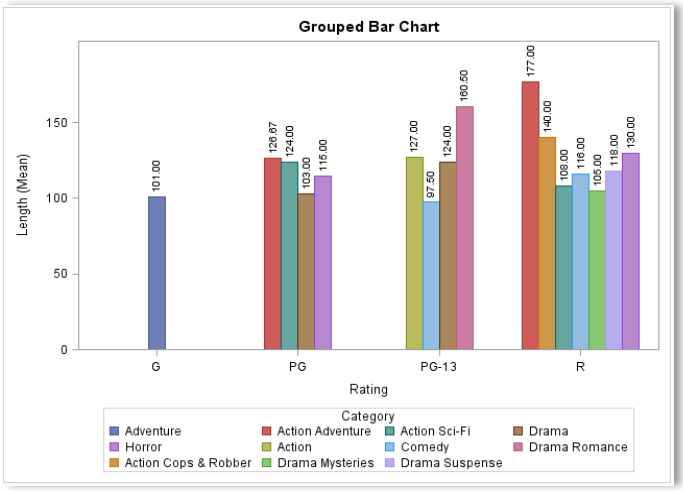
```
TITLE 'Vertical Bar Chart' ;
PROC SGPLOT DATA=MOVIES ;
  VBAR RATING / GROUP=RATING ;
RUN ;
```



Creating a Grouped Vertical Bar Chart

Like the bar chart in the previous example, a grouped bar chart is a commonly used visual for displaying color and comparing the quantity, frequency or other measurements for discrete categories or groups. A grouped bar chart displays the selected variable along the horizontal (x-axis) and uses a grouping variable to display the various sub-category values for each discrete value of the VBAR variable. The next SGPLOT displays the distinct value(s) for the RATING variable and the distinct value(s) for the grouped variable, CATEGORY, along the horizontal axis, and the frequency value along the vertical axis in a VBAR statement. Since the mean value is computed and displayed at the top of each grouped bar for the CATEGORY variable, a FORMAT statement is specified to display values in the desired formatting.

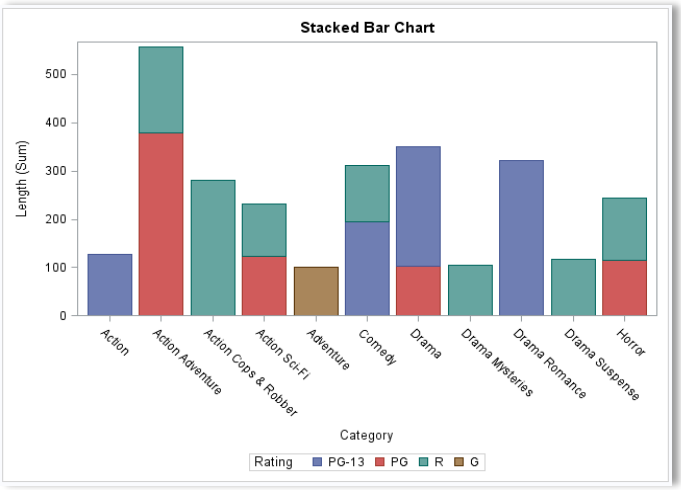
```
TITLE 'Grouped Bar Chart' ;
PROC SGPLOT DATA=MOVIES ;
  FORMAT Length 5.2 ;
  VBAR RATING  /
         RESPONSE=Length
         STAT=Mean
         GROUP=CATEGORY
         GROUPDISPLAY=Cluster
         DATALABEL ;
RUN ;
```



Creating a Stacked Vertical Bar Chart

Like the bar charts in the previous two examples, a stacked bar chart displays and compares the quantity, frequency or some other measurement for discrete categories or groups. A stacked bar chart displays the distinct values for the selected variable of the VBAR statement and stacks the grouping variable, designated with the GROUP= option) along the horizontal (x-axis). The next SGPLOT displays the distinct value(s) for the CATEGORY variable and stacks the distinct value(s) for the grouped variable, RATING, along the horizontal axis, and the frequency value along the vertical axis in a VBAR statement.

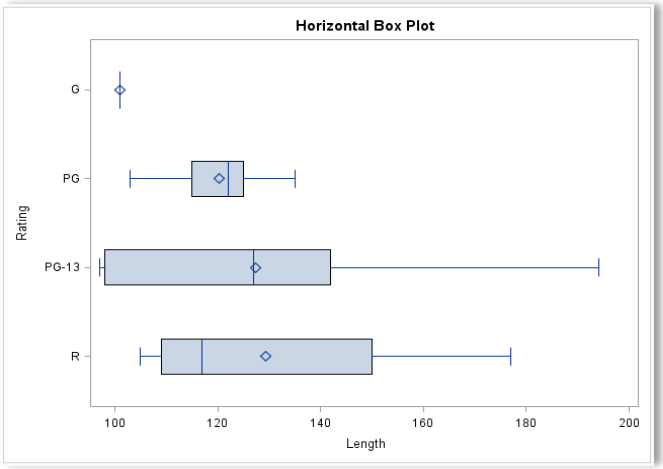
```
TITLE 'Stacked Bar Chart' ;
PROC SGPLOT DATA=MOVIES ;
  VBAR CATEGORY /
        RESPONSE=Length
        GROUP=RATING ;
RUN ;
```



Creating a Horizontal Box Plot

Box plots are useful for displaying data outliers and for comparing distributions for continuous variables. Box plots split the data into quartiles where the range of values traverses the first quartile (Q1) through the third quartile (Q3). The median of the data is represented by a vertical line drawn in the box at the Q2 quartile. Box plots also display the range (or spread) of the data indicating the distance between the smallest and largest value. The next SGPLOT shows the LENGTH variable in an HBOX statement with the RATING variable specified in the CATEGORY= option. The LENGTH variable is displayed on the horizontal (x-axis) and the categorical variable, RATING, is displayed on the vertical (y-axis) of the horizontal box plot output.

```
TITLE 'Horizontal Box Plot' ;
PROC SGPLOT DATA=MOVIES ;
  HBOX Length /
        CATEGORY=Rating ;
RUN ;
```

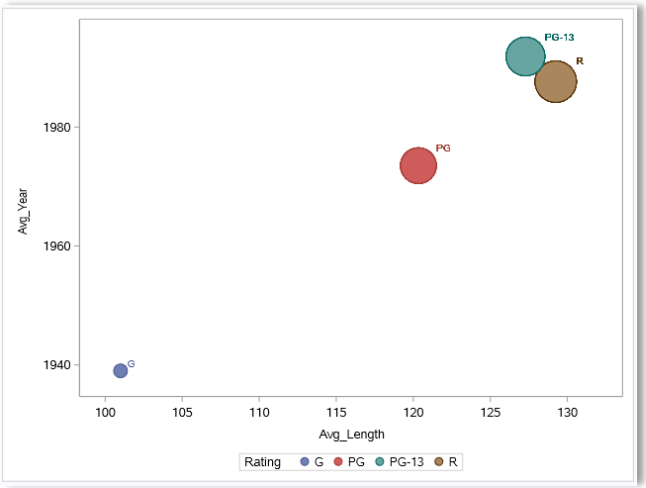


Creating a Bubble Plot

Bubble plots provide a three-dimensional way to visualize data. Considered as a variation of the scatter plot, a bubble plot replaces the dots with bubbles providing an interesting way to view a set of values in three-dimensions. Visuals are created using a horizontal and vertical axis for the purpose of determining whether a possible relationship exists between one variable and another, along with a third variable to determine the size of each bubble. The relationship between two variables is often referred to as their correlation. Bubble plot output is produced with the SGPLOT procedure and a BUBBLE statement. In the following example, the resulting bubble plot contains the data points for the average LENGTH variable on the horizontal (x-axis) and the average YEAR variable on the vertical (y-axis) along with the number of Movies in each Rating group from the MOVIES data set.

```
PROC SQL ;
  CREATE TABLE Movies_Summary as
    SELECT Title, Rating,
           COUNT(Rating) AS
                 Count_Rating,
           AVG(Length) AS
                 Avg_Length,
           AVG(Year) AS Avg_Year
      FROM Movies
        GROUP BY Rating ;
QUIT ;

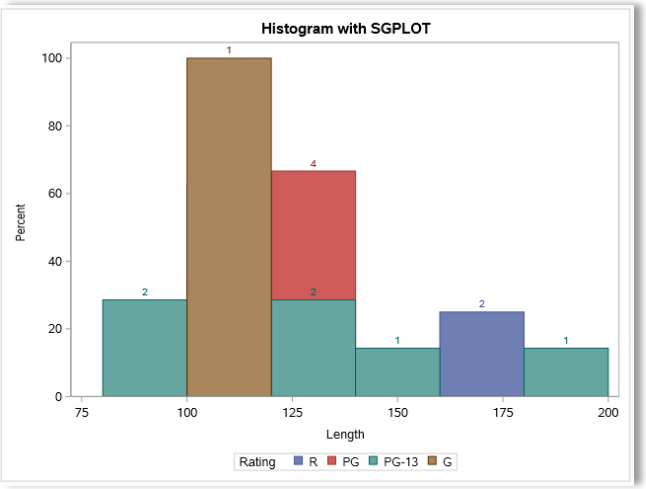
PROC SGPLOT DATA=Movies_Summary ;
  BUBBLE X=Avg_Length Y=Avg_Year
         SIZE=Count_Rating /
               GROUP=Rating
           DATALABEL=Rating ;
RUN ;
```



Creating a Histogram

Histograms are vertical bar charts that display the distribution of a set of numeric data. They are typically used to help organize and display data to gain a better understanding about how much variation the data has. Much can be learned from viewing the shape of a histogram. The various histogram shapes include **Bell-shape** – displays most of the data clustered around the center of the x-axis, **Bimodal** – displays two data values occurring more frequently than any other, **Skewed Right** or **Skewed Left** – displays values that tend to be occurring around the high or low points of the x-axis, **Uniform** – displays data equally (no peaks) across a range of values. The next SGPLOT shows the LENGTH variable specified in a HISTOGRAM statement with the MOVIES data set as input. The LENGTH variable is displayed on the horizontal (x-axis) of the histogram output.

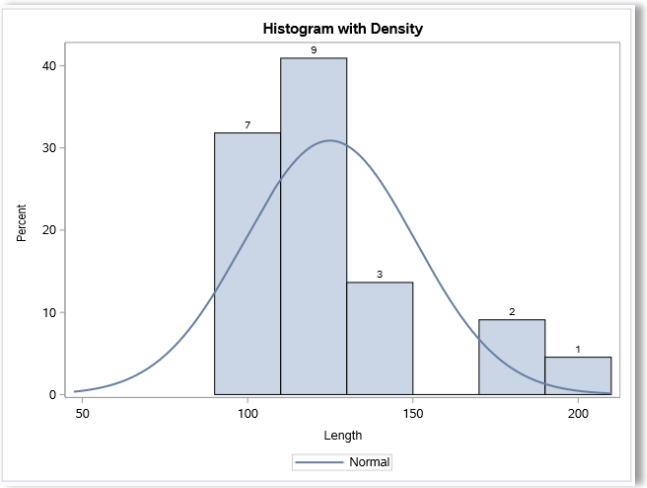
```
TITLE 'Histogram with SGPLOT' ;
PROC SGPLOT DATA=MOVIES ;
  HISTOGRAM Length / GROUP=Rating
                 DATALABEL=Count ;
RUN ;
```



Creating a Histogram with a Density Plot

Histograms can also have a density curve applied to display the distribution of values for numeric data. The next SGPLOT shows the LENGTH variable specified in a HISTOGRAM statement combined with the LENGTH variable displayed in a density plot with the DENSITY statement. As before, the LENGTH variable is displayed on the horizontal (x-axis) of the histogram output.

```
TITLE 'Histogram with Density' ;
PROC SGPLOT DATA=MOVIES ;
  HISTOGRAM Length /
                DATALABEL=Count ;
  DENSITY Length ;
RUN ;
```

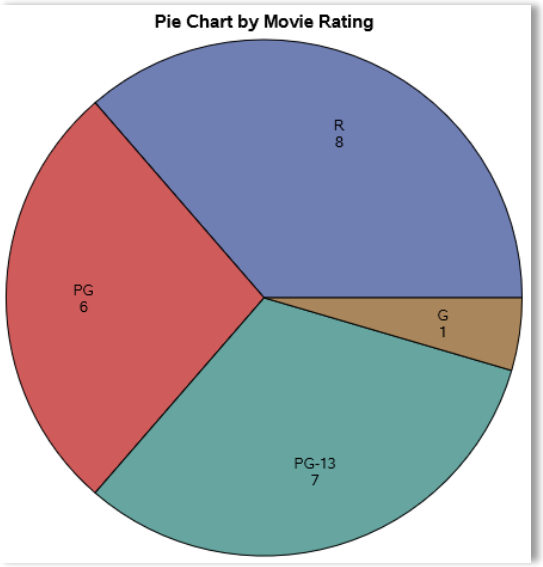


Creating a Pie Chart

Pie charts display data proportionality by using one or more pie slices (maximum of six slices) to illustrate quantity. They are typically used to show the quantity of data categories by displaying pie slices to gain a better understanding of the quantity each category represents. Visualization experts caution users when selecting pie charts because they are difficult to compare data across different pie charts. The following Pie chart template is produced using the Base SAS' TEMPLATE procedure, or statistical graphic language (SGL) along with the SGRENDER procedure. The pie chart shows the quantities of each RATING variable (i.e., "G", "PG", "PG-13", and "R" rated movies) as pie slices.

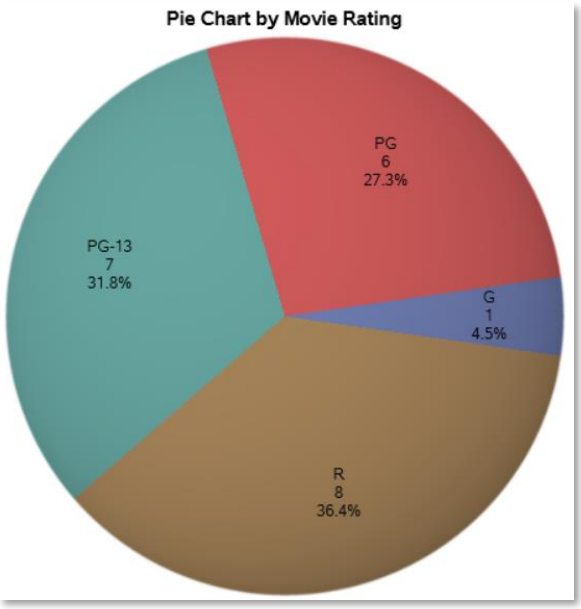
```
proc template ;
  define statgraph PieChart ;
    begingraph ;
    entrytitle "Pie Chart by Movie Rating" ;
      layout gridded / columns=1 ;
        layout region ;
          piechart category=Rating ;
        endlayout ;
      endlayout ;
    endgraph ;
  end ;
run ;
quit ;

proc sgrender data=Movies
              template=PieChart ;
run ;
quit ;
```



Pie charts can also be created using much less code with PROC SGPIE. In the next example, a PROC SGPIE and the PIE statement are specified to show the quantities of each RATING variable (i.e., "G", "PG", "PG-13", and "R" rated movies) as pie slices.

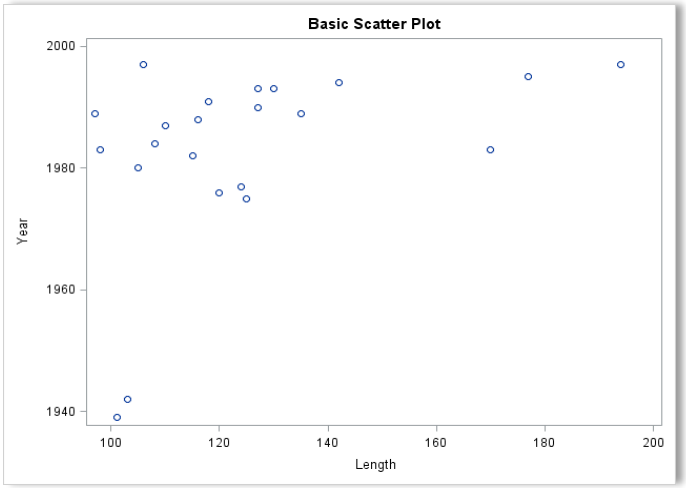
```
proc sgpie data=Movies ;
  title 'Pie Chart by Movie Rating' ;
  pie rating /
        datalabeldisplay=all ;
run ;
```



Creating a Scatter Plot

Scatter plots are typically produced to display data points on a horizontal and vertical axis for the purpose of determining whether a possible relationship exists between one variable and another. The relationship between two variables is often referred to as their correlation. The SCATTER plot output is produced with the SGPLOT procedure and a SCATTER statement. The resulting plot contains the data points for the LENGTH variable on the horizontal (x-axis) and the YEAR variable on the vertical (y-axis) from the MOVIES data set.

```
TITLE 'Basic Scatter Plot' ;
PROC SGPLOT DATA=MOVIES ;
  SCATTER X=Length Y=Year ;
RUN ;
```

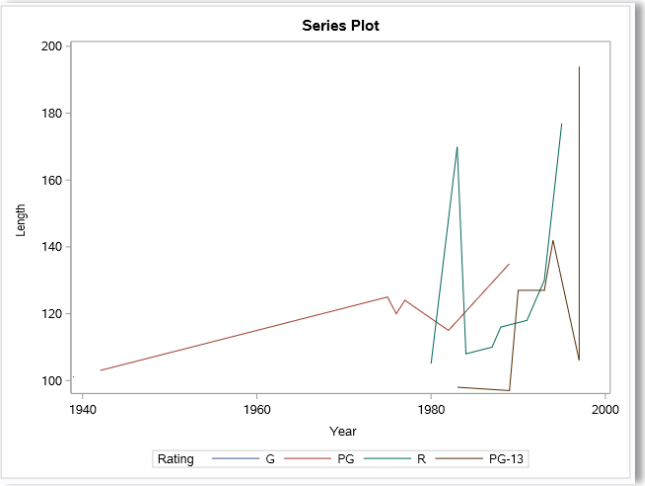


Creating a Series Plot

Series (or Time Series) plots are useful for displaying trends on paired data at different points in time. Series plots display the time variable along the horizontal (x-axis) and data values along the vertical (y-axis). The next SGPLOT selects the YEAR (or time) variable along the horizontal axis and LENGTH along the vertical axis in a SERIES statement.

```
PROC SORT DATA=Movies
          OUT=Sorted_Movies ;
  BY Year ;
RUN ;

TITLE 'Series Plot' ;
PROC SGPLOT DATA=Sorted_Movies ;
  SERIES X=Year Y=Length /
                  GROUP=Rating ;
RUN ;
```



THE SGSCATTER PROCEDURE BY EXAMPLE

Whether the data you work with is large or small, or some size in between, the SGSCATTER procedure is a powerful tool for handling many of your graphical needs. PROC SGSCATTER provides single-statement control to a number of scatter plot panels and matrices. The SGSCATTER procedure supports the following plot statements.

Plot Statement
Plot
Compare
Matrix

The syntax for the SGSCATTER procedure is displayed below.

```
PROC SGSCATTER < DATA=data-set-name > < options > ;

  Plot variable1 * variable2 / < options > ;

  Compare X=(variable(s)) Y=(variable(s)) < / options > ;

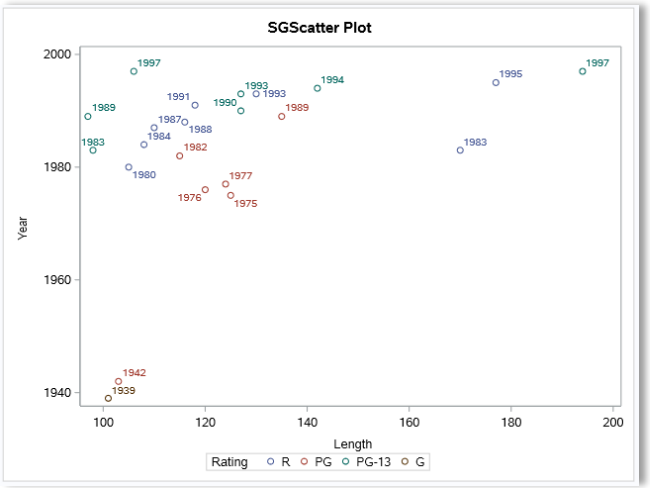
  Matrix variable(s) < / option(s) > ;

RUN ;
```

Creating a Scatter Plot with SGSCATTER

SGScatter plots display data points on a horizontal and vertical axis for the purpose of determining whether a possible relationship exists between one variable and another. The relationship between two variables is often referred to as their correlation. The SCATTER plot output is produced with the SGSCATTER procedure and a PLOT statement. The resulting scatter plot contains the data points for the LENGTH variable on the horizontal (x-axis) and YEAR variable on the vertical (y-axis) from the MOVIES data set.

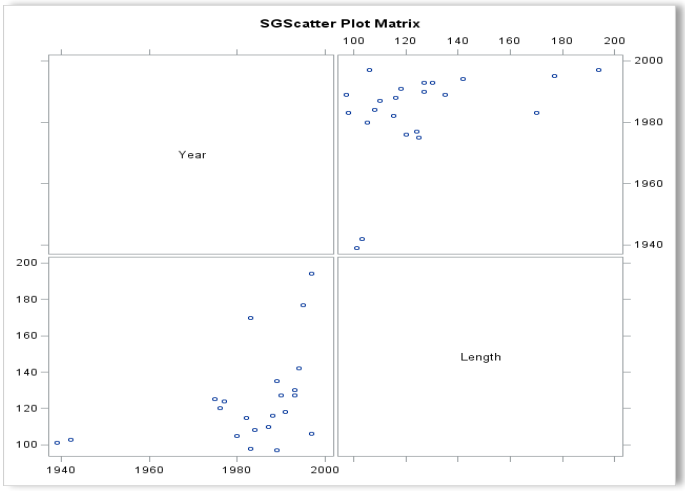
```
TITLE 'SGScatter Plot' ;
PROC SGSCATTER DATA=MOVIES ;
  PLOT Year * Length /
                GROUP=Rating
                DATALABEL ;
RUN ;
```



Creating a Scatter Matrix Plot with SGSCATTER

Scatter matrix plots display data points on a horizontal and vertical axis for the purpose of determining whether a possible relationship exists between one variable and another. The SCATTER plot output is produced with the SGSCATTER procedure and a MATRIX statement. The resulting scatter plot contains a two-pair scatter matrix plot from the MOVIES data set.

```
TITLE 'SGScatter Plot Matrix' ;
PROC SGSCATTER DATA=MOVIES ;
  MATRIX Year Length ;
RUN ;
```



THE SGPANEL PROCEDURE BY EXAMPLE

The SGPANEL procedure provides the ability to create a single-panel of plots or charts using one or more classification variables. Unlike a single-page plot or chart as is produced with the SGPLOT procedure, the SGPANEL procedure is able to produce a panel of plots or charts in a single image, or multiple plots or charts displayed in multiple panels. The result is a panel of plots or charts that display relationships among variables. The SGPANEL supports a number of plot and hart types including histograms, horizontal bar charts, vertical bar charts, horizontal box plots, and vertical box plots. The SGPANEL procedure supports the following statements.

SGPanel Statement
Panelby
Other Plot / Chart Statements

The syntax for the SGPANEL procedure is displayed below.

```
PROC SGPANEL < DATA=data-set >  < options > ;

  PANELBY classvar1 < classvar2 . . . < classvarn > /

              options > ;

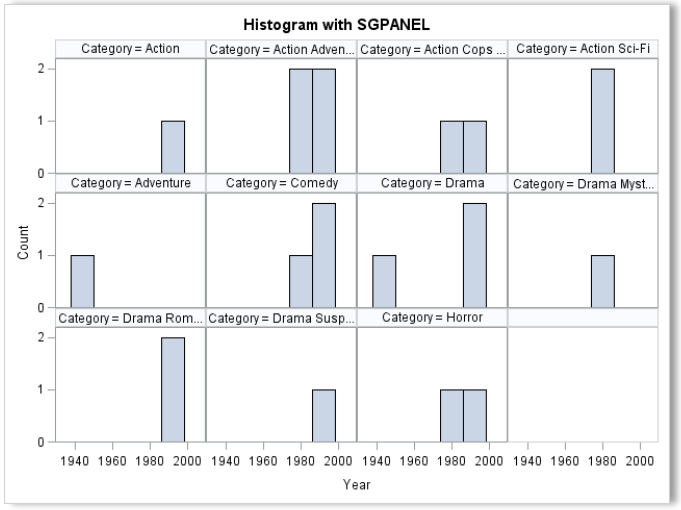
  < plot / chart statement > ;

RUN ;
```

Creating a Panel of Histograms with SGPANEL

A panel of plots and charts created with the SGPANEL procedure displays values of one or more classification variables. The purpose of a panel of plots is to be able to compare one or more variables with one another. The next example shows a panel of histograms created with the SGPANEL procedure and a HISTOGRAM statement. The resulting panel of histograms contains the number of movies (SCALE=Count) by category corresponding to the year the movie was produced from the MOVIES data set.

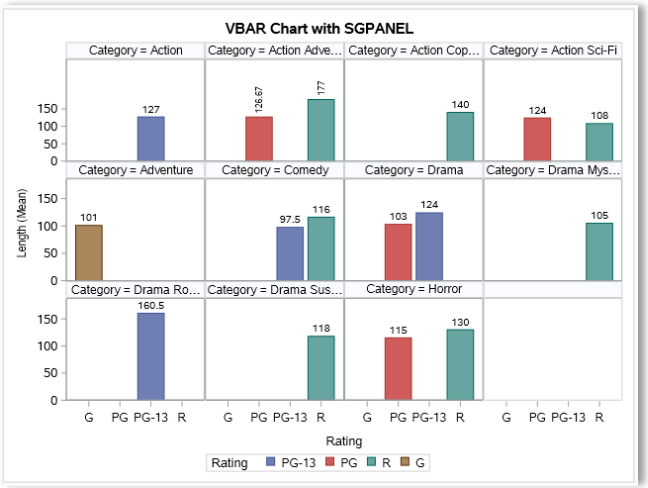
```
PROC SGPANEL DATA=MOVIES ;
  PANELBY CATEGORY / ROWS=3
                     COLUMNS=4 ;
  HISTOGRAM YEAR   / SCALE=COUNT ;
RUN ;
```



Creating a Panel of Bar Charts with SGPANEL

A panel of bar charts can be created with the SGPANEL procedure using one or more classification variables. As before, the purpose of a panel of bar charts could be for comparing one or more variables values against another. The next example shows a panel of vertical bar charts created with the SGPANEL procedure and a VBAR statement. The resulting panel of vertical bar charts contains the average movie lengths (STAT=Mean) for each category of movie corresponding to the movie rating (i.e., G, PG, PG-13, and R) from the MOVIES data set.

```
TITLE 'VBAR Chart with SGPANEL' ;
PROC SGPANEL DATA=MOVIES ;
  PANELBY CATEGORY / ROWS=3
                     COLUMNS=4 ;
  VBAR RATING       / GROUP=Rating
                      RESPONSE=Length
                          STAT=Mean
                      DATALABEL ;
RUN ;
```



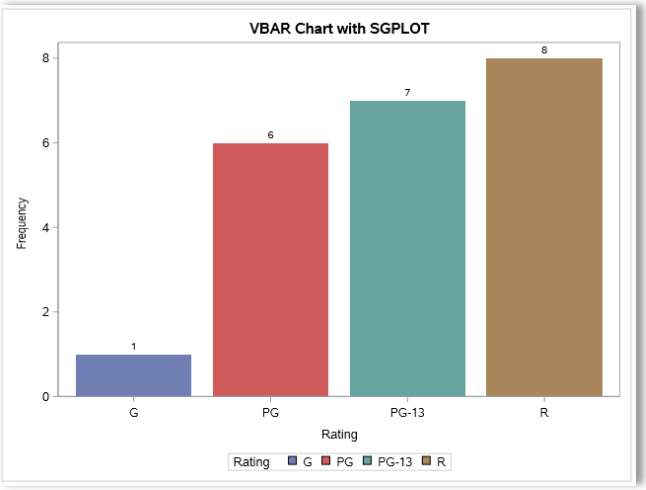
CUSTOMIZING WITH COLOR

The use of color in results, reports, tables, charts, and spreadsheets can help to organize, engage, promote, and encourage greater comprehension with an audience. Color can help make SAS® results, reports, tables, charts and spreadsheets more professional and meaningful. Instead of producing boring and ineffective results, users can enhance the appearance of their results to highlight and draw attention to important data elements and issues, including headings, subheadings, footers, minimum and maximum values, ranges, outliers, special conditions, and other elements. Color can be added to text, foreground, background, row, column, cell, summary, and total with amazing color and traffic lighting scenarios. The following examples illustrate the use of the ODS Statistical Graphic procedures to produce high-quality graphical output for data exploration, data analysis, and statistical analysis.

Using Color in Bar Charts with SGPLOT

Colors and the various color schemes can be applied to chart elements in many ways. Pre-set or fixed colors are available "right-out-of-the-box" with the SGPLOT procedures. Options are available to change the "default" colors and color schemes to address specific needs and requirements. In the next example, the SGPLOT procedure produces a vertical bar (VBAR) chart that uses pre-set "default" colors.

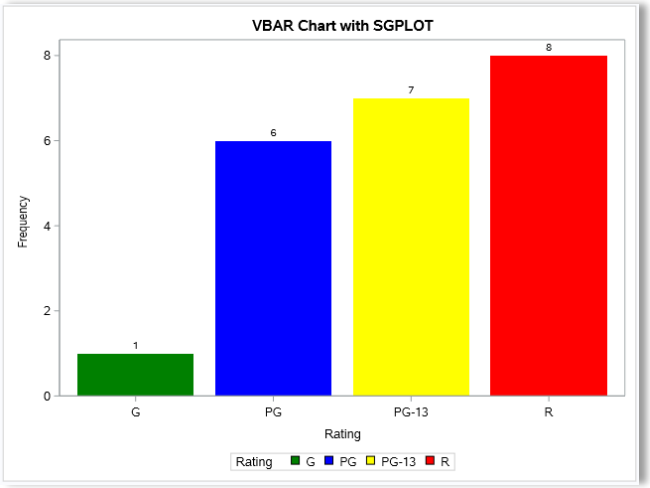
```
TITLE 'VBAR Chart with SGPLOT' ;
PROC SGPLOT DATA=MOVIES ;
  VBAR Rating / GROUP=Rating
                DATALABEL
                FILL
                GROUPDISPLAY=Cluster ;
RUN ;
```



In the next example, the SGPLOT procedure is specified to produce a vertical bar (VBAR) chart with "custom" colors. By defining an attribute map (ATTRMAP) SAS data set with information about the values and colors to use during the creation of the bar chart, users can control which colors, and color schemes, they desire. In producing the vertical bar chart with the SGPLOT procedure, the DATTRMAP= procedure option references the attribute map data set, the GROUPDISPLAY=Cluster and ATTRID=Rating VBAR statement options are specified.

```
DATA ATTRMAP ;
  INPUT @1 ID $6.
        @8 Value $5.
       @14 FillColor $6. ;
  DATALINES ;
Rating G     Green
Rating PG    Blue
Rating PG-13 Yellow
Rating R     Red
;
RUN ;

TITLE 'VBAR Chart with SGPLOT' ;
PROC SGPLOT DATA=MOVIES
        DATTRMAP=ATTRMAP ;
  VBAR Rating / GROUP=Rating
                DATALABEL
                FILL
                GROUPDISPLAY=Cluster
                ATTRID=Rating   ;
RUN ;
```

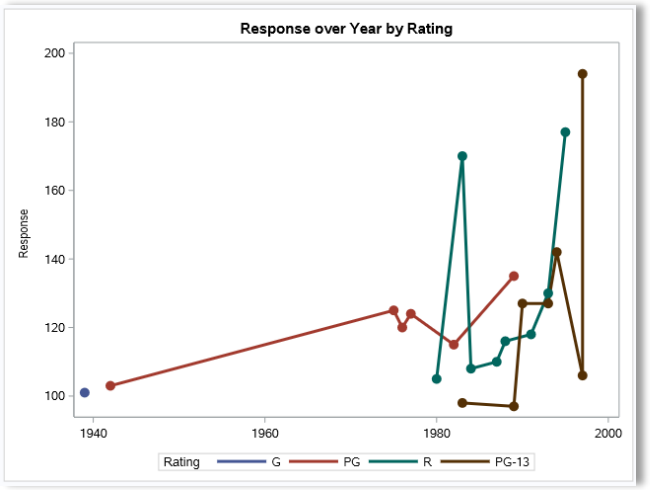


Using Color in Series Plots with SGPLOT

As was shown in the previous section, colors and various color schemes can be applied to plot elements. Pre-set or fixed colors are available “right-out-of-the-box” with the SGPLOT procedures. Options are available to change the “default” colors and color schemes to address specific needs and requirements. In the next example, the SGPLOT procedure produces a series (SERIES) and scatter (SCATTER) plot that uses the pre-set or “default” colors.

```
PROC SORT DATA=mydata.Movies
          OUT=work.Movies_Sorted ;
  BY Year ;
RUN ;

TITLE 'Response over Year by Rating' ;
PROC SGPLOT DATA=work.Movies_Sorted ;
  SERIES X=Year
         Y=Length /
              GROUP=Rating
              LINEATTRS=(THICKNESS=3) ;
  SCATTER X=Year
          Y=Length /
               GROUP=Rating
               MARKERATTRS=
        (SYMBOL=CIRCLEFILLED SIZE=11) ;
  XAXIS DISPLAY=(NOLABEL) ;
  YAXIS LABEL='Response' ;
RUN ;
```

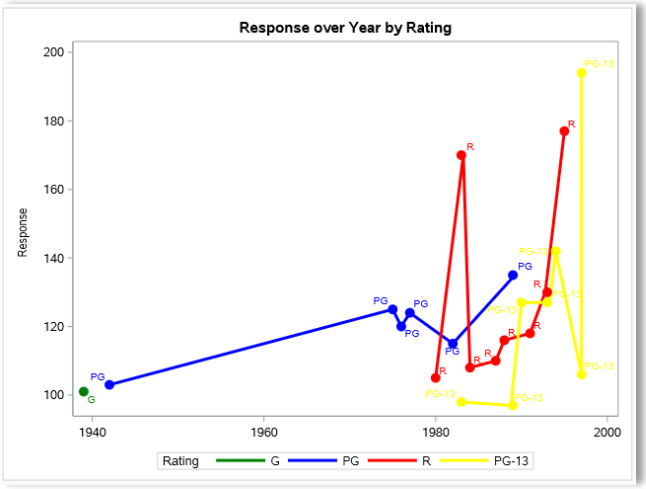


In the next example, the SGPLOT procedure is specified along with a user-defined attribute map (ATTRMAP) to produce a series (SERIES) and scatter (SCATTER) plot with “custom” colors. By defining an attribute map (ATTRMAP) SAS data set with information about the values and colors to use during the creation of the series plot, users can control which colors, and color schemes, they desire. In producing the series plot with the SGPLOT procedure, the DATTRMAP= procedure option references the attribute map data set, the GROUPDISPLAY=Cluster, the ATTRID=Rating, and DATALABEL=Rating SERIES statement options are specified.

```
DATA ATTRMAP ;
  INPUT @1 ID $6.
        @8 Value $5.
       @14 LineColor $6.
       @21 MarkerColor $6. ;
  DATALINES ;
Rating G     Green
Rating PG    Blue
Rating PG-13 Yellow
Rating R     Red
;
RUN ;

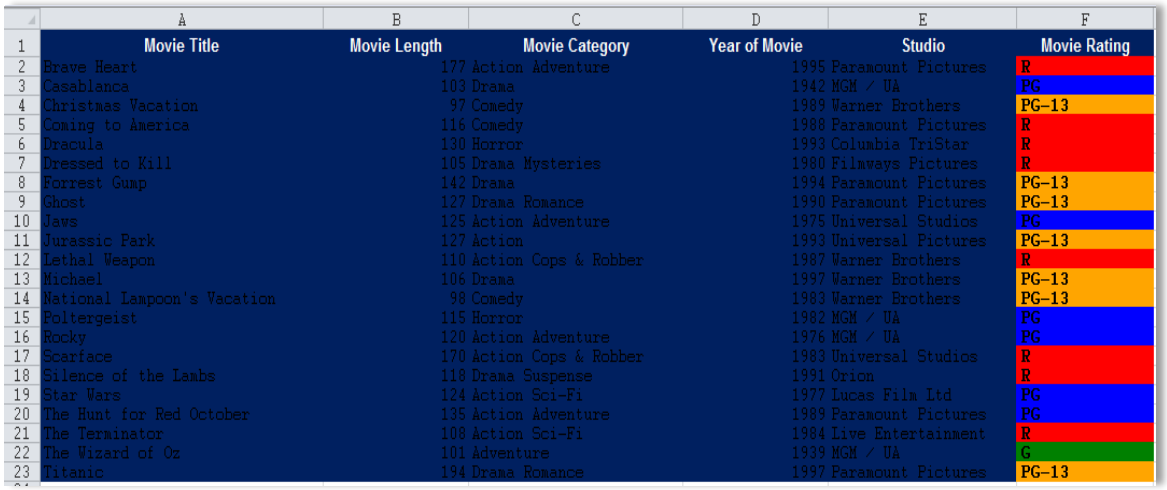
PROC SORT DATA=Movies
          OUT=work.Movies_Sorted ;
  BY Year ;
RUN ;

TITLE 'Response over Year by Rating' ;
PROC SGPLOT DATA=work.Movies_Sorted
        DATTRMAP=ATTRMAP ;
  SERIES X=Year
         Y=Length /
              GROUP=Rating
              LINEATTRS=(THICKNESS=3)
              GROUPDISPLAY=Cluster
              ATTRID=Rating
              DATALABEL=Rating ;
  SCATTER X=Year
          Y=Length /
              GROUP=Rating
              MARKERATTRS=
        (SYMBOL=CIRCLEFILLED SIZE=11)
              ATTRID=Rating ;
  XAXIS DISPLAY=(NOLABEL) ;
  YAXIS LABEL='Response' ;
RUN ;
```



Using Color in Excel Spreadsheets

The use of background and foreground colors in Excel spreadsheets should be chosen carefully and contrast sufficiently with the visual being used. This becomes ever so obvious as we examine the content of an Excel spreadsheet, illustrated in Figure 2, below. As shown, the blue background and black foreground (text) colors make the spreadsheet content difficult to read, particularly because of the insufficient contrast between the black text and blue background colors. Also, the colors used in the traffic lighting for the “Vehicle MSRP” column are difficult to read and distinguish. Color contrast issues should, and must, be avoided to have the results be readable to an audience.



	A	B	C	D	E	F
1	Movie Title	Movie Length	Movie Category	Year of Movie	Studio	Movie Rating
2	Brave Heart	177	Action Adventure	1995	Paramount Pictures	R
3	Casablanca	103	Drama	1942	MGM / UA	PG
4	Christmas Vacation	97	Comedy	1989	Warner Brothers	PG-13
5	Coming to America	116	Comedy	1988	Paramount Pictures	R
6	Dracula	130	Horror	1993	Columbia TriStar	R
7	Dressed to Kill	105	Drama Mysteries	1980	Filmways Pictures	R
8	Forrest Gump	142	Drama	1994	Paramount Pictures	PG-13
9	Ghost	127	Drama Romance	1990	Paramount Pictures	PG-13
10	Jaws	125	Action Adventure	1975	Universal Studios	PG
11	Jurassic Park	127	Action	1993	Universal Pictures	PG-13
12	Lethal Weapon	110	Action Cops & Robber	1987	Warner Brothers	R
13	Michael	106	Drama	1997	Warner Brothers	PG-13
14	National Lampoon's Vacation	98	Comedy	1983	Warner Brothers	PG-13
15	Poltergeist	115	Horror	1982	MGM / UA	PG
16	Rocky	120	Action Adventure	1976	MGM / UA	PG
17	Scarface	170	Action Cops & Robber	1983	Universal Studios	R
18	Silence of the Lambs	118	Drama Suspense	1991	Orion	R
19	Star Wars	124	Action Sci-Fi	1977	Lucas Film Ltd	PG
20	The Hunt for Red October	135	Action Adventure	1989	Paramount Pictures	PG
21	The Terminator	108	Action Sci-Fi	1984	Live Entertainment	R
22	The Wizard of Oz	101	Adventure	1939	MGM / UA	G
23	Titanic	194	Drama Romance	1997	Paramount Pictures	PG-13

Figure 2. Color contrast issues identified in an Excel spreadsheet

By specifying the following PROC FORMAT, PROC REPORT and ODS Excel code to address the color contrast issues identified and shown in the Excel spreadsheet, we're able to make the spreadsheet's content more readable and, as a result, more appealing. The code begins by defining the traffic lighting scenario (i.e., "G-rated" movies are shaded in 'Green', "PG-rated" movies are shaded in 'Light Blue', "PG-13-rated" movies are shaded in 'Orange', and "R-rated" movies are shaded in 'Red') we want using a user-defined format. Then, we specify an ODS Excel statement to create our Excel spreadsheet. Finally, a PROC REPORT is specified to produce the desired layout for the content of our Excel spreadsheet along with the traffic lighting scenario we want displayed for each of the movie ratings. The resulting Excel spreadsheet, shown in Figure 3, illustrates the "default" style, along with the background and foreground color customizations that were specified to resolve the color contrast issues described earlier.

SAS Code to Customize Excel Spreadsheet:

```
PROC FORMAT ;
  Value $RatingFmt
          'G'     = 'Green'
          'PG'    = 'Light Blue'
          'PG-13' = 'Orange'
          'R'     = 'Red' ;
RUN ;

ODS Excel file='D:/Movie Rating with Traffic Lighting.xlsx'
         style=styles.HTMLBlue ;

PROC REPORT DATA=WORK.Movies_Sorted NOWINDOWS
            STYLE(Header)={BackGround=White ForeGround=Black Font=(Calibri,10pt,Bold)} ;
  COLUMNS Title Length Category Year Studio Rating ;
  DEFINE Title    / DISPLAY    'Movie Title'        WIDTH=25 ;
  DEFINE Length   / DISPLAY    'Movie Length'       WIDTH=12  CENTER ;
  DEFINE Category / DISPLAY    'Movie Category'     WIDTH=20 ;
  DEFINE Year     / DISPLAY    'Year of Movie'      WIDTH=13  CENTER ;
  DEFINE Studio   / DISPLAY    'Studio'             WIDTH=25 ;
  DEFINE Rating   / DISPLAY    'Movie Rating'       WIDTH=12  CENTER
         STYLE(Column)=[FontWeight=bold BackGround=$RatingFmt.] ;
RUN ;

ODS Excel close ;
```

Customized Excel Spreadsheet:

	A	B	C	D	E	F
1	Movie Title	Movie Length	Movie Category	Year of Movie	Studio	Movie Rating
2	Brave Heart	177	Action Adventure	1995	Paramount Pictures	R
3	Casablanca	103	Drama	1942	MGM / UA	PG
4	Christmas Vacation	97	Comedy	1989	Warner Brothers	PG-13
5	Coming to America	116	Comedy	1988	Paramount Pictures	R
6	Dracula	130	Horror	1993	Columbia TriStar	R
7	Dressed to Kill	105	Drama Mysteries	1980	Filmways Pictures	R
8	Forrest Gump	142	Drama	1994	Paramount Pictures	PG-13
9	Ghost	127	Drama Romance	1990	Paramount Pictures	PG-13
10	Jaws	125	Action Adventure	1975	Universal Studios	PG
11	Jurassic Park	127	Action	1993	Universal Pictures	PG-13
12	Lethal Weapon	110	Action Cops & Robber	1987	Warner Brothers	R
13	Michael	106	Drama	1997	Warner Brothers	PG-13
14	National Lampoon's Vacation	98	Comedy	1983	Warner Brothers	PG-13
15	Poltergeist	115	Horror	1982	MGM / UA	PG
16	Rocky	120	Action Adventure	1976	MGM / UA	PG
17	Scarface	170	Action Cops & Robber	1983	Universal Studios	R
18	Silence of the Lambs	118	Drama Suspense	1991	Orion	R
19	Star Wars	124	Action Sci-Fi	1977	Lucas Film Ltd	PG
20	The Hunt for Red October	135	Action Adventure	1989	Paramount Pictures	PG
21	The Terminator	108	Action Sci-Fi	1984	Live Entertainment	R
22	The Wizard of Oz	101	Adventure	1939	MGM / UA	G
23	Titanic	194	Drama Romance	1997	Paramount Pictures	PG-13

Figure 3. Color contrast issues addressed in an Excel spreadsheet

CONCLUSION

Data patterns and differences are not always apparent in tables and reports, making visualization an essential complement to traditional communication methods. Effective visual techniques can reveal insights more clearly and support deeper understanding, an outcome valued by data analysts, statisticians, and a wide range of professionals.

With ODS Statistical Graphics, the SAS user community has access to a powerful suite of tools for producing high-quality charts, figures, and visuals to support data exploration, analysis, and statistical modeling. This paper has presented ten practical rules for leveraging ODS Statistical Graphics, its features, capabilities, and syntax, to design clearer, more effective visualizations.

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Widely respected for his ability to translate complex technical concepts into clear, structured, and approachable learning experiences, Kirk's training courses and presentations consistently resonate with audiences ranging from hands-on practitioners to technical leaders. His sessions are known for combining deep technical insight with real-world examples, best practices, and immediately actionable techniques where his technical content doesn't just explain what to do - it shows how and why, empowering participants with greater confidence, capability, and impact.

Kirk partners with organizations across industries to improve analytics capabilities, optimize systems, and enable teams to work more efficiently and confidently. As an educator and author, he is passionate about helping professionals grow their skills, strengthen their problem-solving abilities, and achieve better outcomes through smarter use of data and software. As the author of several books, including PROC SQL: Beyond the Basics Using SAS, Third Edition (SAS Press, 2019), Kirk is a frequent invited speaker, educator, and keynote presenter at conferences, symposiums, and professional summits worldwide, and a recipient of 29 "Best" awards for contributed papers, hands-on workshops (HOWs), and posters.

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